Problem 7.9

```
function outval = faveprompt()
%{
Frank Sucansky AERE 1610 Lab 5 Problem 7.9
Write a function "faveprompt" that prompts the user for his or her favorite
'something'
where 'something' is a string that is passed to the function. For example, it might
prompt
the user for a favorite food, or favorite color. The function will error-check until the
user
enters a string (of any length except 0), and returns that string.
>> outval = faveprompt("food")
What is your favorite food? ice cream
outval =
'ice cream'
>> outval = faveprompt("food")
What is your favorite food?
No, really, What is your favorite food?
%}
something = input("What is your favorite food: ", "s"); %ask for favorite food
if strlength(something) > 0 %error check
outval = something; %set there favorite equal to outval
else
disp("No, really, What is your favorite food: ") %if they enter nothing ask again
end
end
```

Output :

```
faveprompt
What is your favorite food: apples

ans =

 'apples'
>>
>> faveprompt
What is your favorite food:
No, really, What is your favorite food:
>>
```

Problem 7.3

```
% Write a function that prints two random variables 10-30 together
function[result] = getString()
A = randi([10,30], 1, 1)
B = randi([10,30], 1, 1)
result = [A B] % we use this to get both strings next to each other
end
```

Output :

```
String is:>> MyFirstProgram

A =

 23


B =

 12


result =

 23 12


ans =

 23 12
```

Problem 7.13

You would use strlength over length when the number of characters in an array matters more than the number of elements, this could be important when trying to take into account a character limit, as opposed to a word count, in which you would call strlength. One could also use this function to encode data (just for fun), as you would change each character to a corresponding character for the length of all characters.

Output :

N/A

Problem 7.16

```
function [] = filename()
%filename: will prompt the user separately for a filename and extension and will
%create and return a string with the form 'filename.ext'.
name = input('Enter the filename: ', 's'); %enter file name as str
extension = input('Enter the file extension: ', 's'); %enter extension as str
file = strcat(name, '.', extension); %set file as output, equal to joined string
name,extension (e.g. if name = run and extension = .pdf file = run.pdf
sprintf(file) %print file
end %end function
```

Output :

```
>> filename
Enter the filename: run
Enter the file extension: pdf

ans =

    'run.pdf'

>>
```

Problem 8.3

```
clear,clc
%{
Frank Sucansky AERE 1610 Lab 5 Problem 8.3
Create three cell array variables that store people's names, verbs, and nouns. For example,
names = {'Harry', 'Xavier', 'Sue'};
verbs = {'loves', 'eats'};
nouns = {'baseballs', 'rocks', 'sushi'};
Write a script that will initialize these cell arrays, and then print sentences using one
random element from each cell array (e.g. 'Xavier eats sushi').
%}

names = {"Nathaniel", "Frank", "Luis"};
verbs = {"loves", "eats"};
nouns = {"basketball", "rocks", "pizza"};
```

```
sentence = sprintf('%s %s %s.', names{randi([1,3])}, verbs{randi([1,2])},
nouns{randi([1,3])}); %create sentence of random words
disp(sentence); %display Sentence
```

Output :

Frank eats pizza.

Problem 8.9

```
phone = struct('area','803','loc','878','num','9876');
```